left multi-function switch and the left or right turn signal inputs to the instrument cluster, but in parallel with the other turn signal circuits. This arrangement allows the turn signal indicators to remain functional, regardless of the condition of the other circuits in the turn signal and hazard warning systems. The turn signal switch and hazard warning switch inputs to the instrument cluster can be diagnosed using conventional diagnostic tools and methods. (Refer to 8 - ELECTRICAL/LAMPS/LIGHTING - EXTERIOR/LEFT MULTI-FUNCTION SWITCH - OPERATION) for more information on the turn signal switch and the hazard warning switch.

VOLTAGE GAUGE

DESCRIPTION

Fig. 28 Battery Charging Condition Icon

A voltage gauge is standard equipment on all instrument clusters. The voltage gauge is located in the upper left quadrant of the instrument cluster, above the fuel gauge. The voltage gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from 9 volts to 19 volts. An International Control and Display Symbol icon for “Battery Charging Condition” is located on the cluster overlay, directly above the hub of the gauge needle (Fig. 28). The voltage gauge graphics are white and blue against a black field except for a single red graduation at each end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The voltage gauge is serviced as a unit with the instrument cluster.

OPERATION

The voltage gauge gives an indication to the vehicle operator of the electrical system voltage. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The voltage gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the left end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **System Voltage Message** - Each time the cluster receives a system voltage message from the PCM indicating the system voltage is between about 9.5 volts and about 18.5 volts, the gauge needle is moved to the relative voltage position on the gauge scale.
- **System Voltage Low (Charge Fail) Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is less than about 9.0 volts (charge fail condition), the gauge needle is moved to the 9 volt graduation on the far left end of the gauge scale and the check gauges indicator is illuminated. The gauge needle remains on the 9 volt graduation and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is greater than about 12.0 volts (but less than about 16.0 volts), or until the ignition switch is turned to the Off position, whichever occurs first.
- **System Voltage High Message** - Each time the cluster receives a message from the PCM indicating the electrical system voltage is greater than about 19.0 volts, the gauge needle is moved to the 19 volt graduation on the far right end of the gauge scale and the check gauges indicator is illuminated. The gauge needle remains on the right end of the gauge scale and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating the electrical system voltage is less than about 16.0 (but greater than about 9.5 volts), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Communication Error** - If the cluster fails to receive a system voltage message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, it will move the gauge needle to the far left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence in order to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the system voltage to control the generator output. The PCM then sends the proper system voltage messages to the instrument cluster. For further diagnosis of the voltage